

Direct-Sum Row-Energy Envelope and the Cauchy Bottleneck

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Abstract

The transition-native twin-prime program contains a complete-period direct-sum row energy which remains after TPC-215 controls reduced-frequency clustering. This paper gives a deterministic bound for that quantity on the literal V46 scales. Writing the emitter as a sum over primes, the condition $4Q < H$ makes the integer cutoff injective modulo each divisor for every fixed prime. The fixed-prime row norm is therefore exact, and one Cauchy inequality across the prime shell yields

$$L^{-1} \mathcal{E}_{\text{dir}} \ll_{\psi} (Q^3/H)(\log U)^3 = x^{11/32+o(1)}.$$

No prime-number theorem and no Mobius cancellation are used. An exact rational fixture with all prime rows supported on the same two residues shows that the shell Cauchy factor cannot be replaced by free orthogonality. The result is a complete-period structural envelope only: the finite physical-window Gram, prime-shell reassembly, arithmetic L^2 , and the twin-prime endpoint remain open.

Claim level. PROVED_STRUCTURAL_L1 / DIRECT_SUM_ROW_ENERGY_ENVELOPE.

The exponent 11/32 is an unconditional source-locked envelope. The finite fixture is exact rational structural QA. No arithmetic saving is claimed.

1 Introduction

The V46 transition scalar uses the literal coefficient

$$c_d = \frac{\mu(d) \log d}{d} \tag{1}$$

on a squarefree divisor band. Its reciprocal emitter is coupled across divisors through common rational frequencies. TPC-213 identified this common physical source, and TPC-214 reduced the complete-period Gram to reduced-denominator clusters [2]. TPC-215 then proved that the cluster Gram is at most an $O((\log x)^2)$ multiple of the divisor direct-sum energy [3].

The next question is a narrower one: how large can that direct-sum row energy be before finite-window cross frequencies are restored? The answer here is a source-locked envelope, not a cancellation theorem. The fixed-prime integer cutoff has a useful exact feature. Since the shell is contained in $(Q, 2Q]$ and $4Q < H$ for large x , distinct admissible integers cannot occupy the same residue modulo a fixed divisor. This removes cross terms within one prime row. Cauchy is used only once, across the prime shell, and the resulting divisor sum is elementary.

There is also a sharp warning. The prime rows need not be orthogonal to one another. A finite exact fixture with primes congruent to one modulo five makes all rows occupy the same two residues. Thus the shell Cauchy step is a genuine bottleneck of this structural proof.

2 Literal source lock

We retain the V46 scales exactly as frozen in the transition compiler [1]:

$$H = x^{21/32}, \quad Q = x^{1/3}, \quad Y_0 = \frac{H}{4Q}, \quad U = x^{133/400}. \quad (2)$$

Let

$$\mathcal{Q}_x = \{q \text{ prime} : Q < q \leq 2Q\}, \quad P = |\mathcal{Q}_x|, \quad (3)$$

and

$$\mathcal{D}_x = \{d \in \mathbb{Z} : Y_0 < d \leq U, \mu(d)^2 = 1\}. \quad (4)$$

For sufficiently large x , the source inequalities give $U < Q$, so every shell prime is invertible modulo every $d \in \mathcal{D}_x$. For a bounded profile ψ , set

$$B_d(r) = \sum_{q \in \mathcal{Q}_x} \sum_{0 < |m| \leq \lfloor dq/H \rfloor} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{m\bar{q} \equiv r \pmod{d}}. \quad (5)$$

Write $M_\psi = \sup_t |\psi(t)|$ and let $\|v\|_{2,d}^2 = \sum_{r \bmod d} |v(r)|^2$.

The exact exponent ledger is

$$\frac{H}{4Q} = \frac{1}{4} x^{31/96} \longrightarrow \infty, \quad \frac{U}{Q} = x^{-1/1200} \longrightarrow 0. \quad (6)$$

Consequently, the proof below works in the asymptotic source range where $4Q < H$. The factor four is important: the shell extends to $2Q$, and fixed-prime injectivity requires $2q < H$, not merely $2Q < H$.

3 Fixed-prime injectivity

For one prime define

$$B_{d,q}(r) = \sum_{0 < |m| \leq n_{d,q}} \psi\left(\frac{Hm}{dq}\right) \mathbf{1}_{m\bar{q} \equiv r \pmod{d}}, \quad n_{d,q} = \left\lfloor \frac{dq}{H} \right\rfloor. \quad (7)$$

Theorem 3.1 (fixed-q no-collision). *For $d \in \mathcal{D}_x$ and $q \in \mathcal{Q}_x$, distinct admissible integers m in (7) give distinct residues modulo d . Hence*

$$\|B_{d,q}\|_{2,d}^2 = \sum_{0 < |m| \leq n_{d,q}} \left| \psi\left(\frac{Hm}{dq}\right) \right|^2 \leq 2M_\psi^2 \frac{dq}{H}. \quad (8)$$

Proof. Suppose $m_1 \neq m_2$ collide. Since q is a unit modulo d , $d \mid (m_1 - m_2)$. On the other hand, $q \leq 2Q$ and (6) give

$$0 < |m_1 - m_2| \leq 2 \left\lfloor \frac{dq}{H} \right\rfloor \leq \frac{2dq}{H} < d.$$

This is impossible. Thus the atom supports are distinct, so the squared norm is the sum of squared atom weights. There are at most $2\lfloor dq/H \rfloor \leq 2dq/H$ atoms, proving (8). $\square$

4 The direct-sum envelope

Theorem 4.1 (direct-sum row-energy envelope). *Let*

$$\mathcal{E}_{\text{dir}} = L \sum_{d \in \mathcal{D}_x} |c_d|^2 \|B_d\|_{2,d}^2, \quad (9)$$

where L is a complete common period. Then

$$\frac{\mathcal{E}_{\text{dir}}}{L} \leq 16M_\psi^2 \frac{Q^3}{H} \sum_{Y_0 < d \leq U} \frac{\mu(d)^2 (\log d)^2}{d} \ll_\psi \frac{Q^3}{H} (\log U)^3. \quad (10)$$

In particular,

$$\frac{\mathcal{E}_{\text{dir}}}{L} \ll_\psi x^{11/32} (\log x)^3, \quad (11)$$

The right-hand scale is $x^{11/32+o(1)}$, but no matching lower bound is claimed.

Proof. Since $B_d = \sum_{q \in \mathcal{Q}_x} B_{d,q}$, Cauchy in the P shell coordinates and Theorem 3.1 give

$$\|B_d\|_{2,d}^2 \leq P \sum_{q \in \mathcal{Q}_x} \|B_{d,q}\|_{2,d}^2 \quad (12)$$

$$\leq \frac{2M_\psi^2 P d}{H} \sum_{q \in \mathcal{Q}_x} q \leq \frac{4M_\psi^2 P^2 d Q}{H}. \quad (13)$$

The interval $(Q, 2Q]$ contains at most $2Q$ integers for $Q \geq 1$, so $P \leq 2Q$. Therefore

$$\|B_d\|_{2,d}^2 \leq 16M_\psi^2 \frac{dQ^3}{H}.$$

Using $|c_d|^2 = \mu(d)^2 (\log d)^2 / d^2$ in (9) proves the first inequality in (10). Finally, comparison with the integral of $(\log t)^2 / t$ gives

$$\sum_{1 \leq d \leq U} \frac{(\log d)^2}{d} \ll (\log(2U))^3.$$

Since $Q^3/H = x^{1-21/32} = x^{11/32}$ and $\log(2U) = O(\log x)$, the result follows. $\square$

Remark 4.2 (claim boundary). The proof uses absolute values in both the fixed-row bound and shell Cauchy. It does not use the signs of $\mu(d)$, any prime cancellation, or the four-packet signed reassembly. Also, $\mathcal{E}_{\text{dir}}$ is a complete-period quantity. The finite physical interval has an off-frequency Gram that is not controlled by Theorem 4.1.

5 Exact aligned-shell adversary

The shell Cauchy step cannot be replaced by a direct orthogonal sum. Consider the finite rational fixture

$$d = 5, \quad H = 500, \quad Q_{\text{scale}} = 100, \quad \{q\} = \{101, 131, 151, 181\}, \quad \psi(t) = \frac{1}{(1+t^2)^2}. \quad (14)$$

Each displayed q is prime and congruent to one modulo five. Moreover, $[5q/500] = 1$, so the only nonzero integers are $m = \pm 1$. Since $q^{-1} = 1 \pmod{5}$, every fixed- q row has support exactly $\{1, 4\}$. The exact rational certificate in the accompanying repository records the individual rows, the combined row, and all norms without floating point arithmetic.

For orientation, the exact values have decimal displays

quantity	decimal display
sum of individual row norms	1.604830605122346
combined row norm	5.946998427183525
combined/direct ratio	3.705686075652919

The decimal values are only presentations of exact rationals. The ratio is strictly larger than one, so positive cross terms occur. The fixture is a finite structural adversary, not a model for an asymptotic prime shell.

Proposition 5.1 (aligned-shell obstruction). *No proof that uses only the individual fixed- q row norms may replace $\left\|\sum_q B_{d,q}\right\|_2^2$ by $\sum_q \|B_{d,q}\|_2^2$ for the literal row family. In particular, free shell orthogonality is refuted in this scoped structural sense.*

Proof. The fixture (14) has common support and an exact combined norm strictly larger than the individual norm sum. Therefore the proposed orthogonal replacement fails on an admissible finite reciprocal-emitter configuration. The statement is scoped to the replacement rule; it does not assert an asymptotic lower bound for the V46 object. $\square$

6 Route evaluation and open gates

Theorem 4.1 is the next quantitative envelope after TPC-215:

$$\text{complete-period direct row energy} \leq x^{11/32+o(1)}. \quad (15)$$

It removes the need to leave this direct-sum object completely unbounded, but it does not provide a saving relative to a required target scale. The exact remaining gates are:

1. attach the normalized complete-period envelope to the literal finite physical window and its off-frequency Gram;
2. preserve the Mobius signs, prime-only shell, block tails, and four-packet reassembly in one theorem;
3. pay the arithmetic L^2 , fixed-atom, and strict $1/400$ ledgers.

Route A is not applicable to this analytic twin-prime session. The current Route-B structural threshold passes, while arithmetic advance is NO, $L^2 = \text{NONE}$, fixed-atom credit is zero, and full Gate B is open.

7 Conclusion

The source inequality $4Q < H$ turns each fixed-prime reciprocal emitter row into an exact sum of distinct integer atoms. Shell Cauchy and the elementary Mobius-log divisor sum then give the complete-period envelope $L^{-1}\mathcal{E}_{\text{dir}} \ll_{\psi} x^{11/32}(\log x)^3$. The aligned rational fixture shows why the shell Cauchy factor is a real bottleneck rather than a notation artifact. The next research step is a finite-window attachment that keeps this alignment issue and the literal signed reassembly visible.

References

- [1] Liang Wang. Bridge a / gates a–b v46: Transition-native euler carrier and the ap–bdh residual gate, 2026. Internal research artifact, `prime_dynamics_theory` repository.
- [2] Liang Wang. Mobius-weighted shared-frequency clusters in the physical cross-divisor gram, 2026. TPC-214 project, `prime_dynamics_theory` repository.
- [3] Liang Wang. Short-quotient mobius tails and a no-power-loss majorant for the physical cluster gram, 2026. TPC-215 project, `prime_dynamics_theory` repository.